

Progressive Layer Activation CLIP for Few-Shot and Generalizable Cassava Disease Recognition

Muhammad Shafay, Muhammad Owais, Divya Velayudhan, Taimur Hassan, Irfan
Hussain, Naoufel Werghi

KHALIFA UNIVERSITY & ABU DHABI UNIVERSITY, UAE



2026 18TH INTERNATIONAL CONFERENCE ON COMPUTER AND AUTOMATION ENGINEERING (ICCAE)

Motivation

- **The Global Challenge:**

- Cassava is a key food crop for more than 500 million people worldwide.
- These diseases cause **over \$1 billion in losses** every year.



- **The Problem:**

- Expert disease diagnosis **is slow and hard to scale**.
- Traditional AI needs large labeled datasets.



From Manual Inspection to Deep Learning

• Traditional Diagnosis

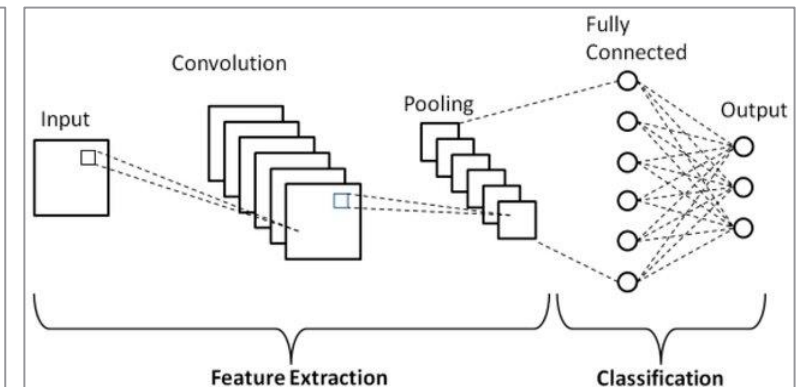
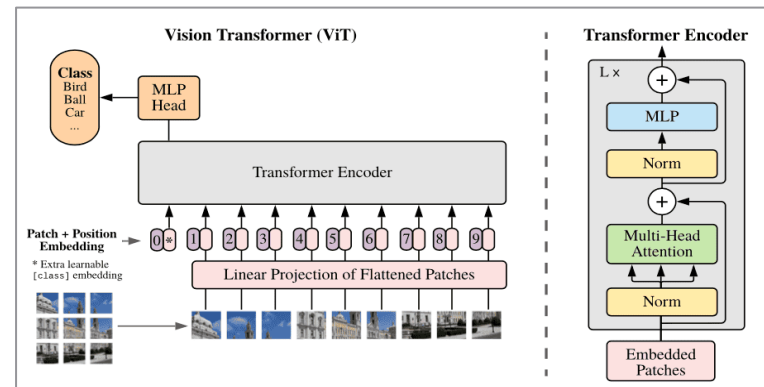
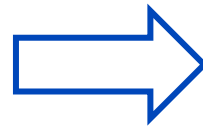
• Manual Expert Inspection

- Visual leaf examination
- Subjective interpretation
- **Labor-intensive and time-consuming**
- Difficult to scale across large farms

• AI-Based Diagnosis

• Deep Learning-Based Systems

- CNNs and Vision Transformers
- Automated feature extraction
- Near-perfect benchmark performance
- **High throughput**



Traditional AI vs. Vision-Language Models (VLMs)

- **Traditional AI Performance:**

- Conventional computer vision models can achieve competitive accuracy when trained on **limited few-shot data** within a specific domain.
- **Underfitting and overfitting** >> poor performance in **real-field conditions**.

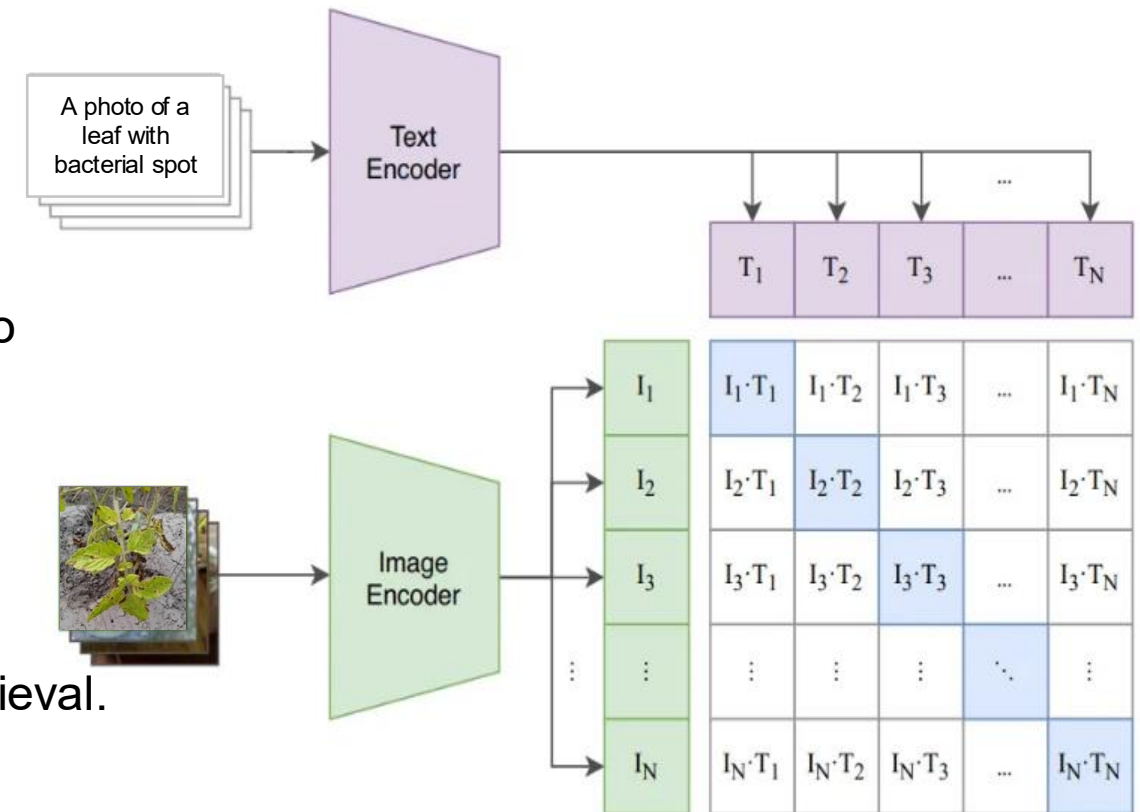
- **VLM Generalization:**

- Vision-Language Models (VLMs) like CLIP **leverage massive pretraining** to maintain a superior global understanding of visual and textual concepts.

Contrastive Language-Image Pre-training (CLIP)

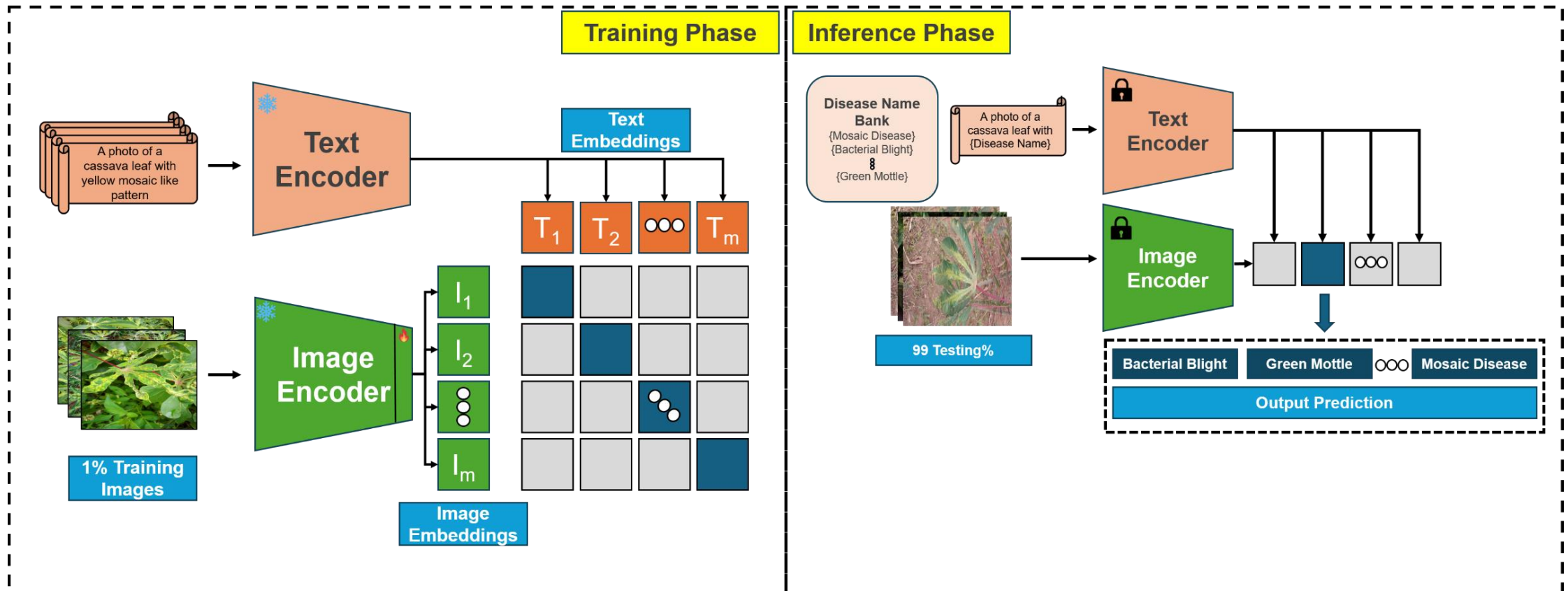
A Vision–Language Foundation for Generalization

- CLIP: links images with texts by using contrastive learning to match images with their descriptions.
- Dual encoders: one for images, one for text.
- Enables tasks like image recognition, classification, and retrieval.



Proposed Method: PLA-CLIP

- **Core Concept:**
 - Progressive Layer Activation (PLA).
- **Strategy:**
 - Instead of unfreezing the whole model at once, **PLA-CLIP gradually unfreezes** transformer layers in the vision encoder.



Training Strategy & Configuration

Few-Shot setup:

- Trained using only 1% of the dataset (43 images per class).

$$\text{ActiveLayers}(s) = \begin{cases} \{T_9 - T_{11}\}, & s = 1, \\ \{T_6 - T_{11}\}, & s = 2, \\ \{T_3 - T_{11}\}, & s = 3, \\ \{T_0 - T_{11}\}, & s = 4, \end{cases}$$

Unfreezing Schedule:

- 4-stage process over 50 epochs.

Three distinct datasets used to test robustness and generalization:

CASSAVA DATASETS USED FOR PLA-CLIP EVALUATION.		
Dataset	Classes	Usage
CD1	CBB, CBSD, CGM, CMD, Healthy	Train (1%) + Test (99%)
CD2	CBSD, CMD, Healthy	Cross-test
CD3	CBB, CMD, Healthy	Cross-test

Results: Classification Performance

- PLA-CLIP outperforms both traditional CV and standard CLIP fine-tuning
- Gain:
 - +15.98% over zero-shot CLIP and +3.94% over standard fine-tuning of CLIP.
 - +4.29% over the best performing Vision Centric Baseline (ViT-Base)

Method	Accuracy (%)	F1-weighted (%)
<i>Vision–Language Approaches</i>		
Zero-shot CLIP [9]	61.27	48.83
CoOp [17]	58.60	62.13
CLIP-Adapter [18]	59.40	56.47
FT-CLIP	74.31	75.75
<i>Traditional Computer Vision</i>		
ResNet50 [25]	65.40	67.03
ViT-Base [26]	73.96	74.16
Swin Transformer [27]	72.28	73.46
MobileNet [28]	51.43	55.85
EfficientNet [29]	43.97	47.84
<i>Proposed Method</i>		
PLA-CLIP (Ours)	78.25	78.00

Results: Cross-Dataset Generalization

- **Robustness:**
 - When tested on unseen datasets (CD2 and CD3) without any additional fine-tuning, PLA-CLIP maintained superior performance.
 - CD2 Performance:
 - 75.85% accuracy (vs. 54.09% for base CLIP).
 - CD3 Performance: 52.63% accuracy (vs. 37.28% for base CLIP).
- **Conclusion:**
- **The progressive strategy improves the transferability of visual-textual representations to different geographic regions.**

Dataset	Model	Accuracy (%)	F1-weighted (%)
CD2	Zero-shot CLIP	54.09	38.59
CD2	PLA-CLIP (Ours)	75.85	81.50
CD3	Zero-shot	37.28	21.69
CD3	PLA-CLIP (Ours)	52.63	62.11

Ablation: Schedule Sensitivity Analysis

- **Best Configuration:** 4-stage schedule achieved the highest performance (78.25% accuracy, 78.0% F1).
- **Strategy Effect:** Performance gain comes from the progressive activation strategy, not hyperparameter tuning.
- **Overfitting Risk:** 5-stage schedule caused performance drop due to excessive unfreezing and overfitting.

Schedule	Accuracy (%)	F1-weighted (%)
3 stage	72.93	74.84
4 stage	78.25	78.00
5 stage	67.29	70.39

Interpretability: Attention Visualizations

Green Mottle



Healthy



Bacterial Blight



Bacterial Streak



Bacterial Blight



Mosaic Disease



Model Efficiency & Inference

- **Real-time Readiness:**
 - Achieves an inference time of **2.65 ms**.
- **Balance:**
 - While larger than **MobileNet (570.91 MB)**, it offers a much-needed accuracy boost for critical plant pathology tasks.
- **Speed:**
 - Operates faster than **Swin-Base (5.36 ms)** while achieving higher accuracy.

Model	Params (M)	GFLOPs	Inference (ms)	Size (MB)
ResNet50	25.56	4.11	1.41	97.80
ViT-Base	86.57	16.87	2.37	330.29
Swin-Base	87.77	15.47	5.36	334.94
MobileNetV2	3.50	0.31	1.29	13.61
EfficientNet-B0	5.29	0.40	1.99	20.46
PLA-CLIP (Ours)	149.62	16.87	2.65	570.91

Conclusion & Future Work

- **Summary:**

- PLA-CLIP provides a **stable, data-efficient way** to adapt large models to specialized **agricultural** domains.

- **Impact:**

- Enables **high-accuracy** disease monitoring even when **labeled data is scarce**.

- **Future Directions:**

- Developing **adaptive (automated) unfreezing** strategies.
- Extending the framework to **other crops**.
- Integrating **temporal and environmental data** for more efficient predictive monitoring.

Thank You